UNIT 7: ORGANISMS – ANIMAL SYSTEMS
7.9 – SENSE ORGANS

Biology 9 Mr. Queenan

Ch. 26 (pg. 874-875)

Sense Organs Outline

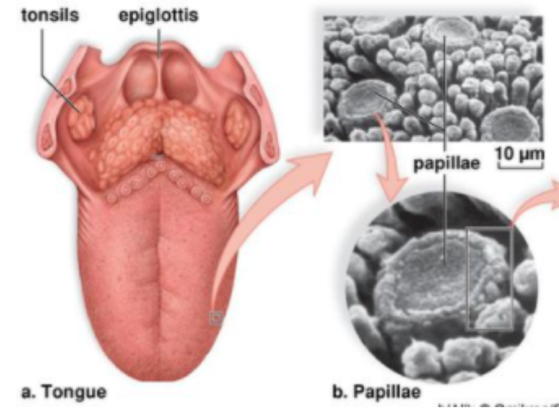
- Sensory Receptors
- Chemical Senses
- Sense of Vision
- Senses of Hearing and Balance
- Somatic Senses

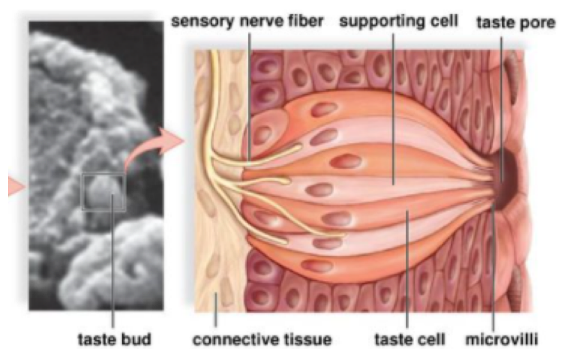
Sensory Receptors

- **Sensory receptors**
  - Specialized cells capable of detecting changes in internal or external conditions, and of communicating that information to the central nervous system
  - Capable of facilitating **sensory transduction**
    - Conversion of an event (stimulus) occurring in the environment into a nerve impulse
    - There is no difference between the nerve impulses carried by different types of sensory nerves
    - **Perceptions** – Any sensory stimuli of which humans, and perhaps other animals, become conscious
  - **Chemoreceptors**
    - Sensory receptors responsible for taste and smell
  - **Photoreceptors**
    - Sensory receptors responsible for responding to light
  - **Mechanoreceptors**
    - Sensory receptors stimulated by mechanical forces, such as pressure
  - **Thermoreceptors**
    - Sensory receptors stimulated by changes in temperature

Chemical Senses

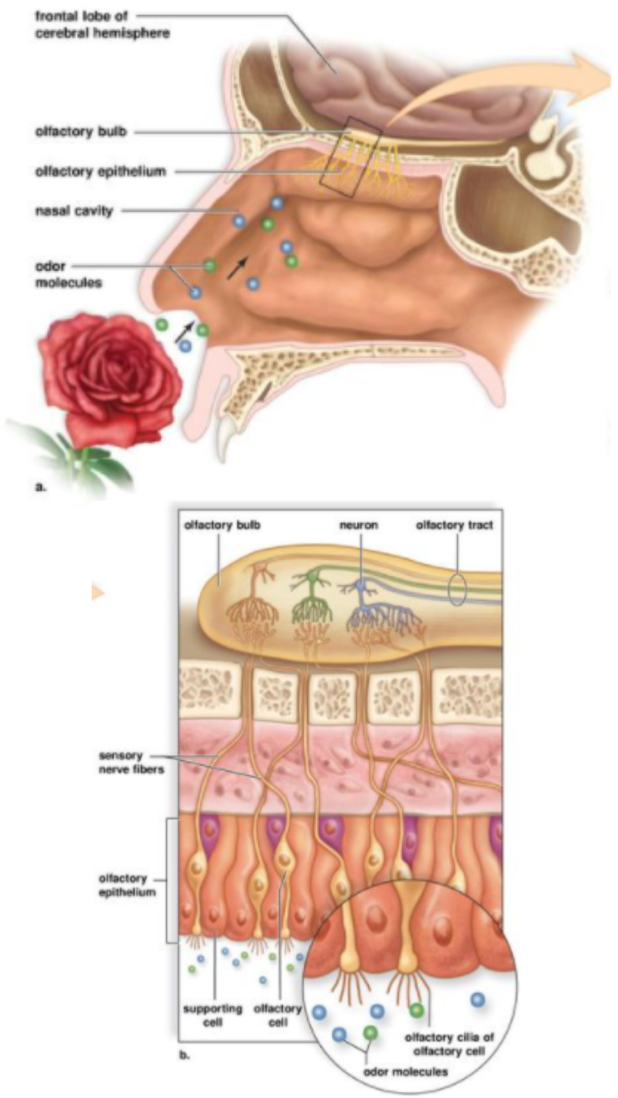
- Chemoreception is found almost universally in all animals
  - Thought to be the most primitive sense
  - Allow organisms to locate food, find a mate, detect dangerous environmental chemicals, etc.
- Sense of Taste in Humans
  - In humans, **taste buds** are located primarily on the tongue
    - Taste buds open at a taste pore
    - Taste buds have supporting cells and elongated taste cells that end in microvilli
    - Five primary tastes
      - Sweet, sour, salty, bitter, and umami (savory)
      - Taste buds for each are located throughout the tongue, although certain regions may be more sensitive to particular tastes





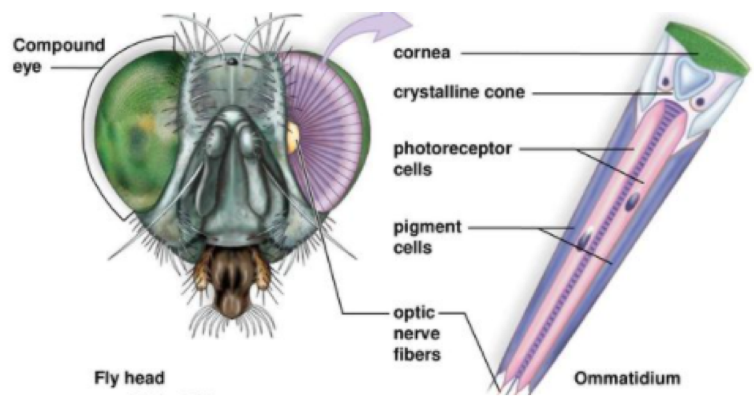
Chemical Senses

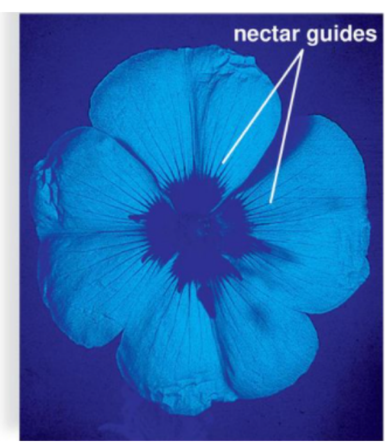
- Sense of Smell in Humans
  - Dependent on olfactory cells
    - Located within olfactory epithelium in the roof of the nasal cavity
  - Nerve fibers from olfactory cells lead to the same neuron in the olfactory bulb
  - Sense of taste and smell
    - Work together to create a combined effect
    - Interpreted by the cerebral cortex



Sense of Vision

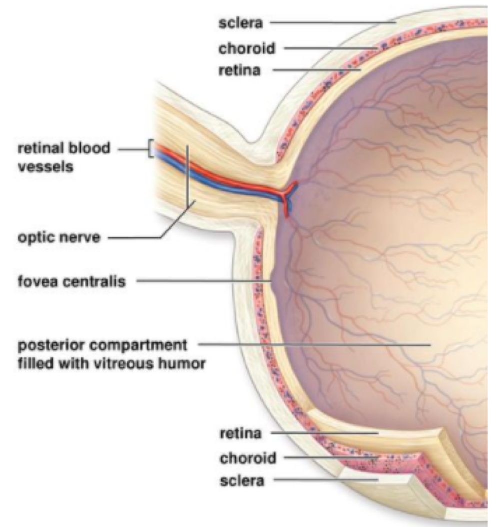
- How Animals Detect Light
  - **Photoreceptors** are sensory receptors that are sensitive to light
  - Arthropods
    - Contain **compound eyes** composed of many independent ommatidia
    - Photoreceptors generate nerve impulses which pass to the brain by way of optic nerve fibers
    - Insects have limited color vision, some able to see UV

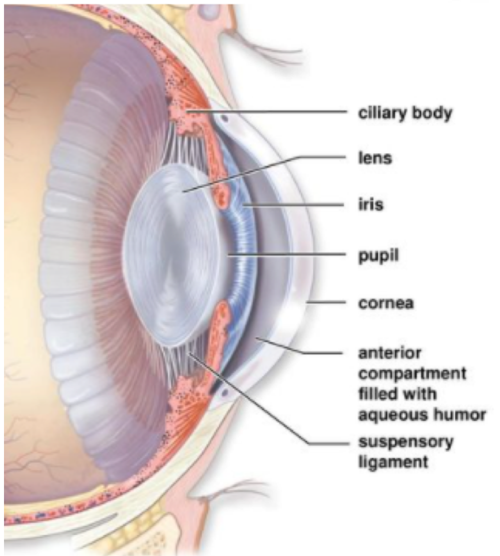




Sense of Vision

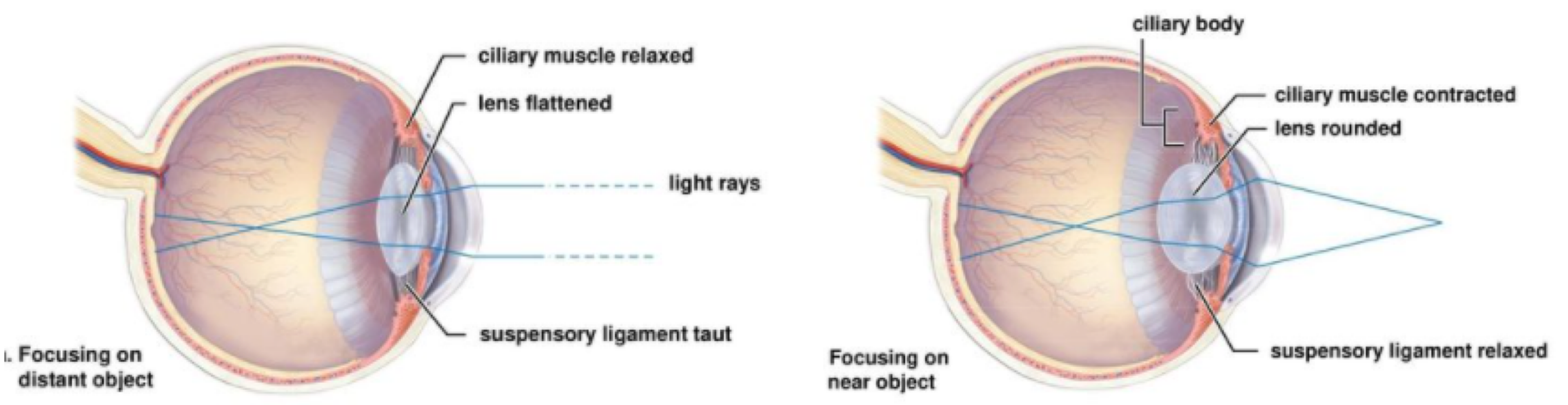
- The Human Eye
  - Three Layers
    - **Sclera** - Opaque outer layer
      - Fibrous layer covering most of the eye
      - In front of the eye, the sclera becomes the transparent **cornea**
      - **Conjunctiva** – covers surface of the sclera and keeps the eyes moist
    - **Choroid** - Thin middle layer
      - Contains blood vessels
      - In front of the eye, the choroid thickens to form the ciliary body and the **iris**
      - The iris regulates the size of the **pupil**
      - The **lens** helps form images
    - **Retina** - Inner layer
      - Contains photoreceptors called **rod cells** and **cone cells**
      - Contains the **fovea centralis**
      - Region of densely packed cone cells where light is focused
        - Site where optic nerve connects to the eye creates a **blind spot,** as no receptors are present





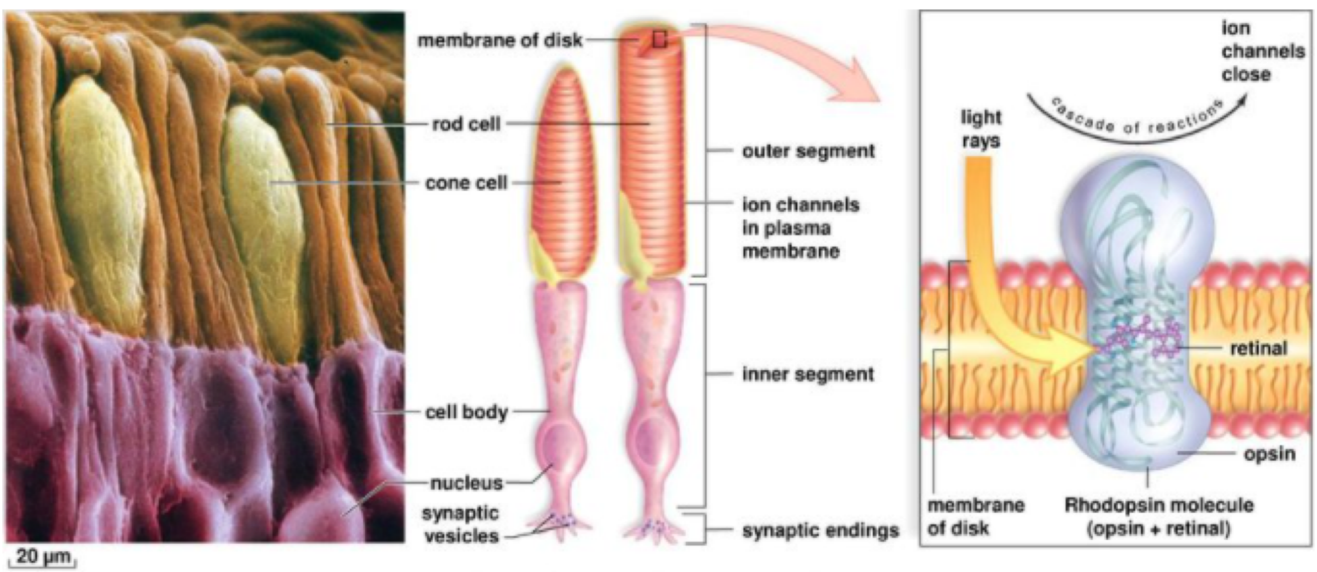
Sense of Vision

- Focusing of the Eye
  - Light rays pass through the pupil and are focused on the retina
  - Focusing starts at the cornea and continues as rays pass through the lens, which provides **visual accommodation**
    - Shape of lens is controlled by the **ciliary muscle**
      - Distant object – ciliary muscle is relaxed
      - Near object – ciliary muscle is contracted



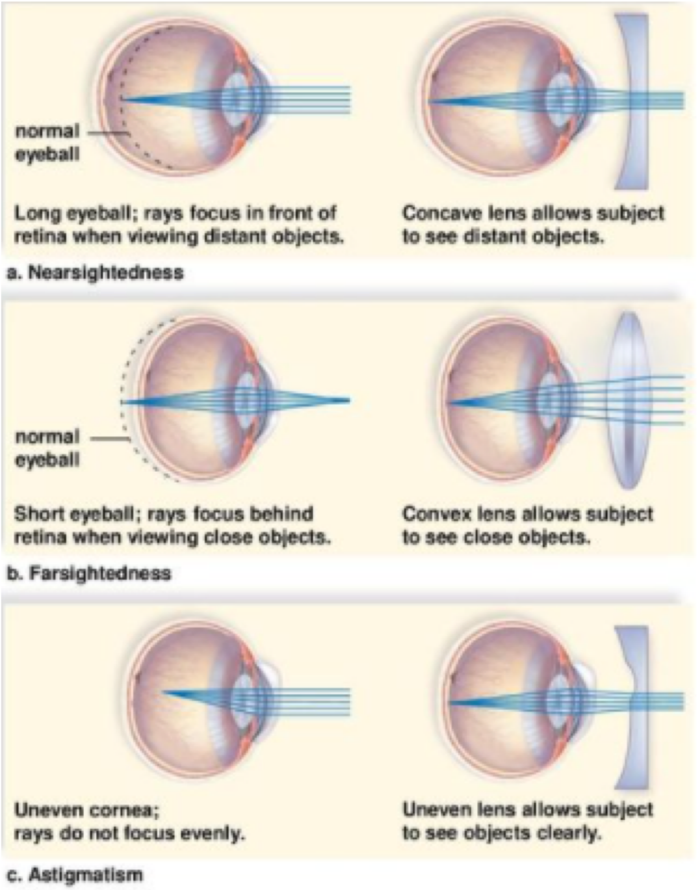
Sense of Vision

- Photoreceptors of the Eye
  - Both rods and cones have an outer segment joined to an inner segment by a stalk
  - Pigment molecules are embedded in membrane of the disks in the outer segment
    - **Rhodopsin** - pigment composed of opsin and retinal, a derivative of vitamin A
  - Rods permit vision in low light
    - Peripheral vision and motion
  - Cones permit vision in bright light
    - Fine detail and color
    - Three different types of cones, each containing a different type of photopigment
      - B (blue), G (green), R (red)



Sense of Vision

- Integration of Visual Signals in the Retina
  - The retina has three layers of neurons
    - Layer closest to choroid contains rod and cone cells
    - Middle layer contains bipolar cells
    - Innermost layer contains ganglion cells
  - Rod and cone cells synapse with the bipolar cells, which in turn synapse with ganglion cells responsible for initiating nerve impulses
  - Human retina has approximately
    - 150 million rod cells
    - 6 million cone cells
    - 1 million ganglion cells
  - Since more rod cells synapse on a ganglion cell than cone cells, cone cells provide sharper more detailed images of an object than rod cells

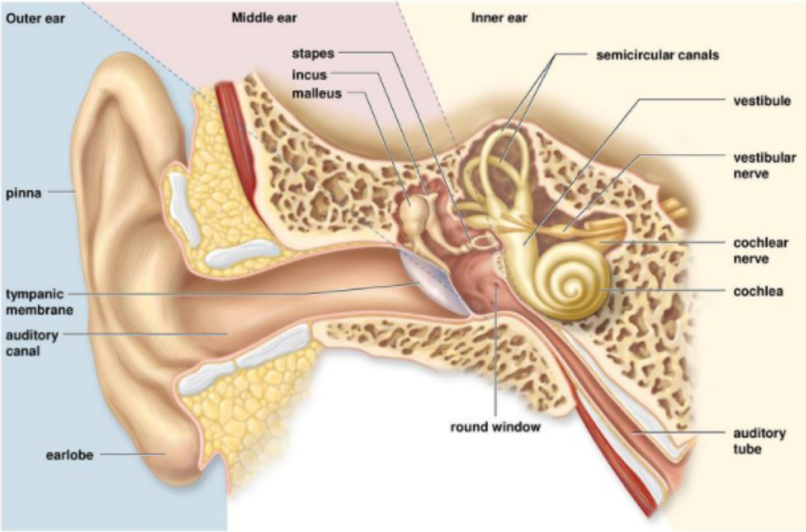


Senses of Hearing and Balance

- How Animals Detect Sound Waves
  - Insects
    - Tympanum – a membrane stimulated to vibrate by sound waves that directly activates nerve impulses in attached receptor cells
  - Fish
    - Lateral line system – detects water currents and pressure waves from nearby objects, helping the organism guide their movement and locate other fish

Senses of Hearing and Balance

- The Human Ear
  - **Outer ear** - Pinna and auditory canal
  - **Middle ear** begins at tympanic membrane and ends at *oval* and *round windows*
    - The **ossicles** (three small bones) are found between the tympanic membrane and the windows
      - *Malleus* (hammer), *incus* (anvil), and *stapes* (stirrup)
    - **Auditory tube** (eustachian tube) extends from the middle ear to the nasopharynx
  - **Inner ear** has three areas
    - **Semicircular canals** - associated with equilibrium
    - **Vestibule** - associated with equilibrium
    - **Cochlea** – associated with hearing



Senses of Hearing and Balance

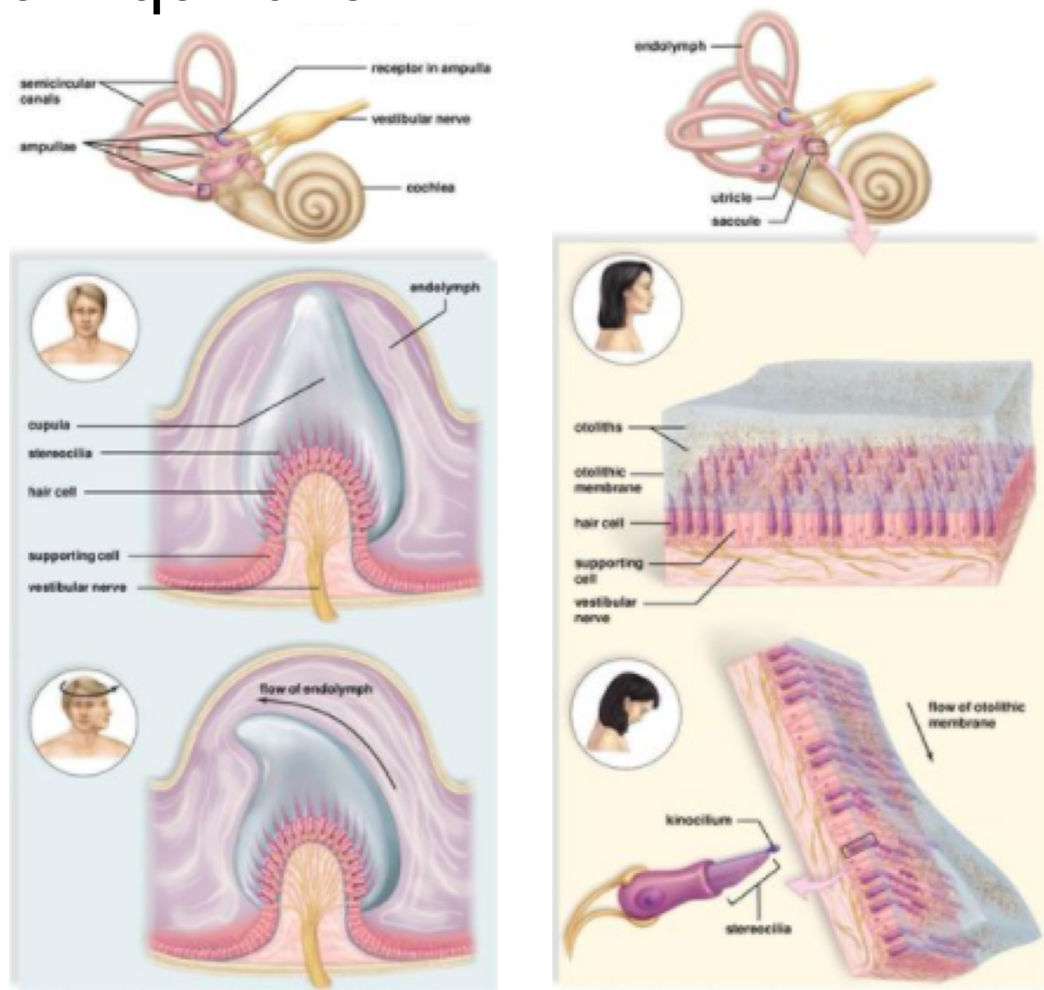
- The Auditory Canal and Middle Ear
  - Sound waves enter the auditory canal
  - Sound waves strike the tympanic membrane, causing it to vibrate
    - Malleus takes pressure from inner surface of tympanic membrane
    - Passes it via the incus to the stapes, multiplying the pressure along the way
    - Stapes strikes the membrane of oval window, passing pressure to fluid within the cochlea
- Inner Ear
  - Spiral organ (organ of Corti) contains hair cells that synapse with nerve fibers of the cochlear nerve
  - Mechanoreceptors for sound are the hair cells of the cochlear canal
    - Sound causes the basilar membrane to vibrate
    - The stereocilia of the hair cells bend
    - Nerve impulses begin in the cochlear nerve and travel to the brain stem, then to the auditory center of the cerebral cortex

Senses of Hearing and Balance

Mechanoreceptors for Equilibrium

- Rotational Equilibrium
  - Ampullae of semi-circular canal contain hairs
  - Shift when rotational force occurs
    - Sends nerve impulse to vestibular nerve
- Gravitational Equilibrium
  - Utricle and saccule contain hair cells
  - When head bends, hairs shift in the membrane fluid
    - Sends nerve impulse to vestibular nerve

The different nerve impulses let the brain know head orientation at all times



a. Rotational equilibrium: receptors in ampullae of semicircular canal

b. Gravitational equilibrium: receptors in utricle and saccule of vestibule

Somatic Senses

- **Somatic senses**
  - Senses whose receptors are associated with the skin, muscles, joints, and viscera
    - **Proprioceptors**
      - Mechanoreceptors involved in reflex actions that maintain muscle tone
        - Muscle spindles, Golgi tendon organs
    - **Cutaneous receptors**
      - Make the skin sensitive to touch, pressure, pain, and temperature
        - Meissner corpuscles
        - Krause end builds
        - Merkel disks
        - Pacinian corpuscles
        - Ruffini endings
    - **Pain Receptors**
      - Also termed free nerve endings or *nociceptors*
      - Damaged cells release chemicals causing nociceptors to generate nerve impulses that are interpreted by the brain as pain.